

Practice Midterm Exam

Answer Key

Total: 100 points

Grading Note: The short-answer rubrics below provide suggested point allocations. Equivalent explanations that demonstrate correct conceptual understanding should receive full credit. Students do not need to use exactly the same wording as the answer key.

Part I. Multiple Choice

40 points

Each question is worth 2 points.

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	C	B	B	B	B	A	C

Question	11	12	13	14	15	16	17	18	19	20
Answer	C	C	B	C	B	A	C	B	B	B

1. Centering

Answer: B. Centering

Subtracting the mean of a variable from each observation centers that variable:

$$x' = x - \bar{x}.$$

The centered variable has mean 0.

2. Autoscaling

Answer: A. Its mean is 0 and its standard deviation is 1.

Autoscaling is

$$z = \frac{x - \bar{x}}{s}.$$

Centering gives a mean of 0, while division by the standard deviation gives a standard deviation of 1.

3. Pareto Scaling

Answer: B. Pareto scaling

Pareto scaling reduces the influence of high-variance variables but preserves more of the original variance structure than autoscaling. It can be written as

$$x' = \frac{x - \bar{x}}{\sqrt{s}}.$$

4. Normalization

Answer: C. Some samples have systematically higher overall signal because different amounts of material were analyzed.

Normalization primarily addresses systematic sample-to-sample differences. Scaling, in contrast, primarily changes the relative weighting of variables.

5. Log Transformation

Answer: B. Reduce skewness and compress large values

A log transformation can reduce strong right skewness and decrease the relative influence of very large measurements. It does not guarantee normality or remove all outliers.

6. Simple Linear Regression

Answer: B. A one-unit increase in X is associated with a 2-unit increase in predicted Y .

The slope describes the expected change in the response associated with a one-unit increase in the predictor. The regression coefficient alone does not establish causation.

7. Residual

Answer: B. $y_i - \hat{y}_i$

A residual is the difference between the observed response and its predicted value:

$$e_i = y_i - \hat{y}_i.$$

8. Multiple Linear Regression

Answer: B. Holding X_2 constant, a one-unit increase in X_1 is associated with a 4-unit increase in predicted Y .

The phrase “holding the other predictor constant” is essential when interpreting a coefficient in multiple regression.

9. Regression Coefficients

Answer: A. Regression coefficients can depend on which other predictors are included in the model.

In multiple regression, a coefficient represents the association between a predictor and the response conditional on the other predictors in the model. Adding another predictor can therefore change a coefficient.

10. Association and Causation

Answer: C. Regression can describe associations, but causal conclusions require additional assumptions and study-design considerations.

A regression coefficient alone does not demonstrate that changing a predictor will cause the response to change.

11. Logistic Regression

Answer: C. Log-odds

Logistic regression models

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

Thus, the linear predictor corresponds to the log-odds of the positive outcome.

12. Probability and Odds

Answer: C. 1

The odds are

$$\frac{p}{1-p} = \frac{0.5}{0.5} = 1.$$

13. Odds Ratio

Answer: B. A one-unit increase in X_1 multiplies the odds of $Y = 1$ by 0.5.

In logistic regression,

$$e^{\beta_1}$$

is the odds ratio associated with a one-unit increase in X_1 , holding other predictors constant.

14. Probability and Classification Threshold

Answer: C. The predicted class depends on the classification threshold.

A logistic regression model produces a predicted probability. Converting that probability into a class label requires choosing a classification threshold.

15. Precision

Answer: B. Precision

Precision is

$$\text{Precision} = \frac{TP}{TP + FP}.$$

It answers: among all observations predicted to be positive, what fraction are actually positive?

16. Recall

Answer: A. Recall

Recall is

$$\text{Recall} = \frac{TP}{TP + FN}.$$

When missing a true disease case is costly, high recall is especially important.

17. Lowering the Threshold

Answer: C. More observations are classified as positive.

Lowering the threshold makes it easier for an observation to be classified as positive. This generally increases both true positives and false positives.

18. AUROC

Answer: B. Performance equivalent to random ranking

An AUROC of approximately 0.5 indicates that the classifier has no discriminative ranking ability beyond random performance.

19. Overfitting

Answer: B. Training accuracy 98%, validation accuracy 68%

The very high training performance combined with substantially poorer validation performance is characteristic of overfitting.

20. Hyperparameter Selection

Answer: B. Validation data or cross-validation within the development data

Hyperparameters should be selected using validation data or cross-validation within the development dataset. The test dataset should remain untouched until final model evaluation.

Part II. Conceptual Understanding

60 points

21. Preprocessing Choices

10 points

(a) Identify the two preprocessing problems.

4 points

Answer:

There are two distinct problems:

1. The metabolites have very different numerical scales or variances. This can cause variables with large magnitudes or variances to dominate some analyses.
2. One batch has systematically higher overall signal intensity because of differences in the amount of sample material injected. This is a sample-to-sample systematic difference.

Suggested grading:

- 2 points: identifies the variable-scale/magnitude problem.
- 2 points: identifies the systematic sample-to-sample intensity difference.

(b) Which problem is addressed by scaling?

2 points

Answer:

Scaling primarily addresses differences among **variables**. It changes their relative weighting so that variables with large scales or variances do not automatically dominate the analysis.

Suggested grading:

- 1 point: identifies the first problem.
- 1 point: explains that scaling changes the relative weighting of variables.

(c) Which problem is addressed by normalization?

2 points

Answer:

Normalization primarily addresses systematic differences among **samples**, such as differences caused by the amount of material injected or overall signal intensity.

Suggested grading:

- 1 point: identifies the second problem.
- 1 point: explains that normalization addresses systematic sample-to-sample differences.

(d) Why can autoscaling make a low-variance metabolite influential?

2 points

Answer:

Autoscaling divides each centered variable by its standard deviation:

$$z = \frac{x - \bar{x}}{s}.$$

If s is very small, dividing by that small value can greatly magnify small fluctuations. If those fluctuations largely represent measurement noise, autoscaling can amplify the influence of noise.

Suggested grading:

- 1 point: recognizes division by a small standard deviation.
- 1 point: explains that this can amplify noise or small fluctuations.

22. Interpreting a Multiple Linear Regression Model

10 points

The model is

$$\widehat{\text{Blood Pressure}} = 120 + 5X_1 - 2X_2.$$

(a) Interpret the coefficient 5.

3 points

Answer:

Holding physical activity (X_2) constant, a one-unit increase in environmental exposure (X_1) is associated with an expected 5-unit increase in predicted blood pressure.

Suggested grading:

- 1 point: one-unit increase in environmental exposure.
- 1 point: 5-unit increase in predicted blood pressure.
- 1 point: holding physical activity constant.

(b) Interpret the coefficient -2 .

3 points

Answer:

Holding environmental exposure (X_1) constant, a one-unit increase in physical activity (X_2) is associated with an expected 2-unit **decrease** in predicted blood pressure.

Suggested grading:

- 1 point: one-unit increase in physical activity.
- 1 point: 2-unit decrease in predicted blood pressure.
- 1 point: holding environmental exposure constant.

(c) Does the coefficient establish causation?

2 points

Answer: No.

The regression coefficient describes an association conditional on the other variables included in the model. It does not by itself demonstrate that increasing exposure will cause blood pressure to increase. Causal interpretation requires additional assumptions and appropriate study design.

Suggested grading:

- 1 point: states that causation is not established.
- 1 point: explains association versus causation or mentions confounding/study design.

(d) Interpret the intercept.

2 points

Answer:

The intercept 120 is the predicted blood pressure when

$$X_1 = 0 \quad \text{and} \quad X_2 = 0.$$

Whether this has a meaningful practical interpretation depends on whether zero values of both predictors are meaningful and plausible.

Suggested grading:

- 1 point: predicted blood pressure when both predictors equal 0.
- 1 point: recognizes that this may or may not be meaningful in the application.

23. Understanding Confounding

8 points

(a) How could age be a confounder?

4 points

Answer:

Age could be a confounder if it is associated with both coffee consumption and the health outcome. For example, coffee consumption may differ across age groups, while age may also independently be associated with the health outcome. Some of the apparent coffee–outcome association in the simple regression could therefore reflect differences in age.

Suggested grading:

- 1 point: age is related to coffee consumption.
- 1 point: age is related to the health outcome.
- 1 point: recognizes that this can distort the apparent coffee–outcome association.
- 1 point: clearly explains the confounding idea.

(b) Why might the coffee coefficient change?

2 points

Answer:

After age is included, the coffee coefficient represents the association between coffee consumption and the outcome **holding age constant**. The model is therefore estimating a different conditional association.

Suggested grading:

- 1 point: coefficient is now adjusted for age.
- 1 point: explains that the coefficient represents a conditional association holding age constant.

(c) Was the simple regression calculated incorrectly?

2 points

Answer: No.

The simple regression can be mathematically correct while answering a different question. Its coefficient describes the unadjusted association between coffee and the outcome, whereas the multiple regression coefficient describes the association conditional on age.

Suggested grading:

- 1 point: states no.
- 1 point: distinguishes unadjusted from age-adjusted association.

24. Logistic Regression Interpretation

8 points

The model is

$$\log\left(\frac{p}{1-p}\right) = -1 + \beta_1 X,$$

with

$$e^{\beta_1} = 2.$$

(a) What does p represent?

2 points

Answer:

p is the probability of the positive outcome:

$$p = P(Y = 1 \mid X).$$

Suggested grading:

- 1 point: identifies p as a probability.
- 1 point: specifies the probability of $Y = 1$ given the predictor.

(b) Interpret $e^{\beta_1} = 2$.

3 points

Answer:

A one-unit increase in X multiplies the odds of $Y = 1$ by 2. Equivalently, the odds of the positive outcome are doubled for each one-unit increase in X .

Suggested grading:

- 1 point: one-unit increase in X .
- 1 point: refers specifically to the odds of $Y = 1$.
- 1 point: correctly states that the odds are multiplied by 2.

(c) Why does this not mean the probability doubles?

3 points

Answer:

The quantity e^{β_1} is an **odds ratio**, not a probability ratio. Probability and odds are related by

$$\text{odds} = \frac{p}{1-p},$$

so multiplying the odds by 2 does not generally multiply the probability by 2. The resulting change in probability depends on the starting probability.

Suggested grading:

- 1 point: identifies e^{β_1} as an odds ratio.
- 1 point: distinguishes odds from probability.
- 1 point: recognizes that the change in probability depends on the starting probability.

25. Choosing a Classification Threshold

8 points

(a) More or fewer positive predictions?

2 points

Answer: More patients will generally be classified as positive.

Lowering the threshold from 0.5 to 0.2 means that patients with predicted probabilities between 0.2 and 0.5 will now also be classified as positive.

(b) What happens to recall?

2 points

Answer: Recall will generally increase.

More true disease cases will be classified as positive, reducing the number of false negatives.

$$\text{Recall} = \frac{TP}{TP + FN}.$$

Suggested grading:

- 1 point: recall increases.
- 1 point: explains the increase through more true positives and/or fewer false negatives.

(c) What happens to false positives?

2 points

Answer: The number of false positives will generally increase.

Lowering the threshold causes more negative patients to be classified as positive as well as more positive patients.

(d) Why might this tradeoff be acceptable?

2 points

Answer:

For screening a serious disease, the cost of missing a true disease case may be substantially greater than the cost of sending an additional patient for follow-up testing. Therefore, accepting more false positives may be reasonable in order to achieve higher recall and fewer false negatives.

Suggested grading:

- 1 point: recognizes that false negatives are particularly costly in this context.
- 1 point: explains why accepting more false positives may be reasonable.

26. Model Complexity and Regularization

8 points

Model	Training Accuracy	Validation Accuracy
A	91%	88%
B	100%	71%

(a) Which model shows stronger evidence of overfitting?

3 points

Answer: Model B.

Model B achieves perfect training accuracy but only 71% validation accuracy. The large training–validation performance gap indicates that the model fits the training data extremely well but does not generalize well to unseen data.

Suggested grading:

- 1 point: identifies Model B.
- 1 point: recognizes the large training–validation gap.
- 1 point: connects the gap to poor generalization/overfitting.

(b) Which model should generalize better?

2 points

Answer: Model A.

Model A has substantially higher validation accuracy (88%) and a much smaller gap between training and validation performance. Based on the available evidence, it is therefore expected to generalize better to unseen data.

Suggested grading:

- 1 point: identifies Model A.
- 1 point: justifies the choice using validation performance and/or the smaller generalization gap.

(c) How might stronger regularization affect Model B?

3 points

Answer:

Stronger regularization penalizes large coefficient values and generally shrinks the coefficients toward zero. This reduces model flexibility or effective complexity and may reduce overfitting. Training performance may decrease, but validation/generalization performance may improve.

Suggested grading:

- 1 point: coefficients are penalized/shrunk.
- 1 point: model complexity or flexibility is reduced.
- 1 point: explains the potential reduction in overfitting and improvement in generalization.

27. Cross-Validation and the Test Set

8 points

(a) How should K -fold CV be used?

3 points

Answer:

Cross-validation should be performed using only the 80% development dataset.

For each candidate value

$$C = 0.01, \quad C = 1, \quad C = 100,$$

the development data are divided into K folds. The model is repeatedly trained on $K - 1$ folds and evaluated on the remaining fold. The cross-validation performance is summarized across folds, and the value of C with the best appropriate validation performance is selected.

Suggested grading:

- 1 point: CV is performed only within the development data.
- 1 point: each candidate C is evaluated across the folds.
- 1 point: the value of C is selected based on CV performance.

(b) What should be done after selecting C ?

2 points

Answer:

After selecting the best value of C , fit the final model using the development data with the selected hyperparameter. Then evaluate that final model on the untouched test dataset.

Suggested grading:

- 1 point: fit/retrain the model using the development data with the selected C .
- 1 point: evaluate the resulting model on the test data.

(c) Why not select C using the test dataset?

3 points

Answer:

If the researcher repeatedly evaluates different values of C on the test dataset and chooses the one with the best test performance, the test data have become part of the model-selection process.

The test dataset is then no longer an independent assessment of generalization performance. The resulting test performance estimate may be optimistically biased.

Suggested grading:

- 1 point: recognizes that the test data would be used for hyperparameter/model selection.
- 1 point: recognizes that the test set would no longer provide an independent final evaluation.
- 1 point: identifies information leakage and/or optimistic bias in the reported performance.

Optional Self-Check Questions — Suggested Answers

These questions are not included in the 100-point score.

1. **What is the conceptual difference between scaling and normalization?**

Answer: Scaling primarily changes the relative weighting of variables, whereas normalization primarily addresses systematic sample-to-sample differences.

2. **What phrase should almost always appear when interpreting a coefficient in multiple linear regression?**

Answer: “Holding the other predictors constant” or an equivalent phrase.

3. **Why is e^{β_j} interpreted as an odds ratio rather than a probability ratio?**

Answer: Logistic regression is linear in the log-odds. Exponentiating a coefficient therefore gives the multiplicative change in odds associated with a one-unit increase in the predictor.

4. **What is the conceptual difference between precision and recall?**

Answer:

$$\text{Precision} = \frac{TP}{TP + FP}$$

asks what fraction of predicted positives are truly positive, whereas

$$\text{Recall} = \frac{TP}{TP + FN}$$

asks what fraction of actual positives are successfully identified.

5. **Why can accuracy be misleading when one class is rare?**

Answer: A classifier can achieve high accuracy simply by predicting the majority class while performing poorly on the rare class.

6. **What happens to recall when the classification threshold is lowered?**

Answer: Recall generally increases because more observations are classified as positive, resulting in more true positives and fewer false negatives.

7. **What pattern in training and validation performance suggests overfitting?**

Answer: Very strong training performance combined with substantially poorer validation performance.

8. **How does decreasing C affect regularization strength in scikit-learn logistic regression?**

Answer: Decreasing C increases regularization strength because C is the inverse of regularization strength.

9. What is an out-of-fold prediction?

Answer: An out-of-fold prediction for a sample is generated by a model that was trained without that sample. In K -fold cross-validation, each sample receives a prediction from the model for which its fold served as the validation fold.

10. Why should the final test dataset remain untouched until model development is complete?

Answer: The test dataset should provide an independent, unbiased assessment of the final model's generalization performance. Using it for model or hyperparameter selection compromises that independence.

End of Answer Key